

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations
 December 2023
BT1102-1- EMERGING BIOPRODUCTS
 (BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Diabetes is treated using**
 A) Insulin
 B) Probiotics
 C) Amino acids
 D) Carbohydrates
- Haemophilus Influenza* is example for _____ vaccine**
 A) Conjugate
 B) Subunit
 C) Recombinant subunit
 D) None from the list
- Alexander Fleming was culturing _____ which is secreting antibiotic**
 A) *Staphylococcus aureus*
 B) *E. coli*
 C) *Penicillium notatum*
 D) None from the list
- _____ is currently blended with petrol.**
 A) Bioethanol
 B) Biodiesel
 C) Biohydrogen
 D) None from the list
- Xanthan gum has the application in _____**
 A) Ice cream preparation
 B) Adhesive manufacturing
 C) Fruit juice clarification
 D) All from the list
- Production of vinegar process involves _____ fermentation step**
 A) Double
 B) Single
 C) Triple
 D) None from the list
- Saccharomyces boulardii* has the application in _____**
 A) Probiotics
 B) Bioethanol
 C) Vaccine
 D) None from the list
- Enzymes are used in the production of _____**
 A) Fructose syrup
 B) Detergent
 C) Dairy products
 D) All from the list
- _____ is example for bioenergy source**
 A) Bioplastic
 B) Biomass
 C) Enzyme
 D) None from the list
- _____ microbial cell-based product is used for domestic application.**
 A) Vinegar
 B) Bioethanol
 C) Amino acids
 D) None from the list
- Bacille Calmette-Guerin* is example for _____ vaccine**
 A) Whole-killed bacterial
 B) Live attenuated and inactivated
 C) Conjugate
 D) None from the list
- Xanthum gum is produced by _____ cell**
 A) Microbial
 B) Animal
 C) Plant
 D) None from the list
- _____ is animal cell based product using hybrid cell**
 A) Monoclonal antibody
 B) Vaccine
 C) Recombinant protein
 D) None from the list

P.T.O.

14. Therapeutic factors are produced using ____ cell
 A) Microbial B) Plant
 C) Animal D) None from the list
15. Biosimilars are commercially produced using ____ cell
 A) Microbial B) Plant
 C) Animal D) All from the list
16. Recombinant proteins are produced using ____ cell using pancreatic genes
 A) Microbial B) Plant
 C) Animal D) All from the list
17. Secondary metabolites are commercially manufactured using ____
 A) Nematodes B) Plant
 C) Human D) All from the list
18. Natural dye are manufactured through ____ cell
 A) Insect B) Plant
 C) Animal D) All from the list
19. ____ is not the example for plant cell -based product.
 A) Bioplastic B) Biobutanol
 C) Probiotics D) All from the list
20. Cosmetics are manufactured using ____ cell
 A) Microbial B) Plant
 C) Animal D) All from the list

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss the overview and application of insulin.	08	L2	1	1,6
b)	Highlight the overview of antibiotics and its applications.	08	L2	2	1,6
2. a)	What steps are integral to the bioethanol production process? Explain with a pertinent flowchart for clarity.	08	L2	1	1,6
b)	What is the definition of single cell proteins, and how are they utilized in various applications?	08	L2	2	1,6
3. a)	What are the primary steps involved in the manufacturing of vinegar? Provide the detailed explanation.	08	L2	3	1,6
b)	List out and explain the application of enzyme.	08	L2	3	1,6
Unit – II					
4. a)	What is the general concept behind producing cosmetics using animal cells?	08	L2	4	1,6
b)	List out and explain applications of biologics.	08	L2	4	1,6
5. a)	Highlight the overview of biosimilars and discuss one application.	08	L2	4	1,6
b)	Explain production of monoclonal antibodies with its essential stages.	08	L2	4	1,6
6. a)	What are recombinant proteins? Highlight the overview and its application.	08	L2	4	1,6
b)	Explain the application of vaccine developed using animal cells.	08	L2	4	1,6
Unit – III					
7. a)	Explain the production process of natural dye.	08	L2	5	1,6
b)	List out and explain the application of algal bioproducts.	08	L2	5	1,6
8. a)	What are secondary metabolites? Explain commercial applications.	08	L2	5	1,6
b)	Explain therapeutics manufactured from plant cells with suitable example	08	L2	5	1,6

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
